

Does random cross-validation overstate the skill of seismic-bump forecasting models?

A leakage study on the UCI Seismic Bumps dataset

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Code, data pipeline and verification script: github.com/junjie-yang11/seismic-bump-forecasting

Abstract

Machine-learning results on the UCI Seismic Bumps dataset — shift-level records from a Polish coal mine, where the task is to forecast whether a hazardous seismic event will occur during the *next* shift — are commonly reported under stratified k -fold cross-validation. Because the records form an ordered sequence and the label is defined relative to the following shift, shuffling rows lets a model train on shifts that occur after the ones it is tested on. This note quantifies the effect.

Under random stratified 10-fold cross-validation a class-weighted logistic regression reaches ROC-AUC 0.755 and PR-AUC 0.195. Under time-ordered expanding-window validation the same model reaches ROC-AUC 0.668 and PR-AUC 0.084 — a 57% relative loss in PR-AUC. A bagged CART ensemble shows the same pattern more strongly (0.222 $\rightarrow$ 0.078, --65%).

Adding the record index — a pure time proxy carrying no physical information — *improves* random-CV scores while *degrading* time-ordered scores, which is the signature of temporal leakage. On a single chronological holdout, both models fall to ROC-AUC 0.568–0.612 with recall 0.115 at an operational threshold. Any claim about the forecasting skill of these models should therefore be based on time-respecting validation.

A final check concerns the probability scale. Class weighting, which is needed here because positives are rare, makes the fit behave as if the base rate were 0.50. The logistic regression’s mean predicted probability is 0.333 against an observed rate of 0.0427; a single log-odds shift reduces the expected calibration error from 0.290 to 0.0144. The same correction over-corrects the tree ensemble, which is already close to calibrated. Ranking metrics are blind to all of this.

1 Introduction

Seismic-bump forecasting is a rare-event prediction problem in which the consequences of a missed event are severe, so the operating point matters more than aggregate accuracy. The dataset’s original study reported results obtained with stratified 10-fold cross-validation repeated ten times, and public reproductions follow the same protocol.

For a task framed as “given the previous shift, will the next shift be hazardous?”, stratified k -fold is not a valid estimate of forecasting skill, because it breaks the temporal order of the sequence. Three distinct mechanisms

can inflate the score: chronological leakage (training rows may post-date test rows); distribution drift (the base rate changes over the record); and duplicate or near-duplicate records. This note isolates the first two on a fixed dataset and two model families, and reports the magnitude of the discrepancy.

2 Data

The dataset comprises 2578 shift records (8 h each) described by 18 monitoring channels, with a binary target. There are 170 positive records, a prevalence of 0.0659. The mirror used here removes six exact duplicates from the 2,584-row original; the positive count is unchanged, so results are comparable in size but not row-for-row with the original file.

2.1 Integrity checks

Table 1. Integrity checks performed before any modelling. Two findings are worth noting: three columns carry no information at all, and the bump-range counts are linearly dependent.

Check	Result	Implication
Duplicate rows	0	none
Constant columns	nbumps6, nbumps7, nbumps89	zero information; add noise dimensions
Rank of nbumps2...nbumps89	4 of 7	exactly 3 linear dependencies, caused by the 3 constant columns
nbumps vs sum of range columns	equal in 2,576 / 2,578 rows	nbumps is a total; dropped to restore full rank

Table 1. Dataset integrity checks. The three constant columns explain the “(3 not defined because of singularities)” message produced by a full logistic regression on all 18 original features.

2.2 Label drift across the record sequence

Block of the record	1	2	3	4	5
P(hazardous shift)	0.1609	0.0680	0.0213	0.0447	0.0349

Table 2. Hazard prevalence in five equal blocks of the record. The first block is $7.6\times$ more hazardous than the third.

Hazard prevalence across the record sequence (5 equal blocks)

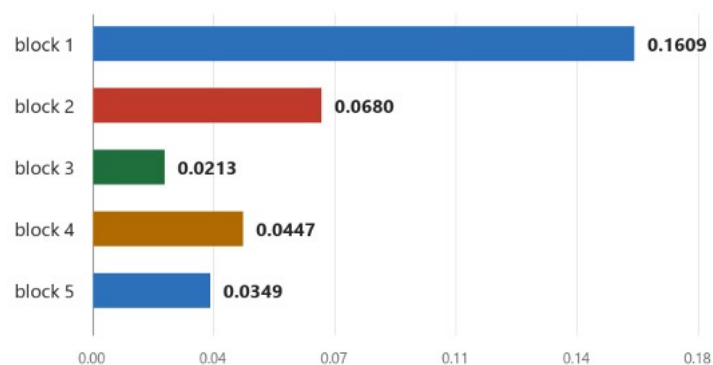


Figure 1. Hazard prevalence across the record sequence. This drift is the structure that random cross-validation exploits.

Whatever its cause — a period of more intense mining, or a change of longwall — this structure means that a random split mixes high- and low-hazard regimes between training and test sets, while a time-ordered split does not.

3 Methods

3.1 Models

Both models are implemented from scratch in NumPy so that every reported number is traceable to a formula:

- **Logistic regression**, L2-regularised ($\lambda = 1.0$ on standardised features), fitted by iteratively reweighted least squares with class weights $n/(2 \cdot n_{\text{pos}})$ and $n/(2 \cdot n_{\text{neg}})$; the intercept is unpenalised.
- **Bagged CART**, 60 bootstrap trees, depth 6, minimum 20 samples per leaf, Gini impurity, candidate splits at 12 quantiles per feature. Included so that the conclusion is not an artefact of a linear model.

Feature standardisation is fitted inside each training fold only. Scaling on the full dataset before cross-validation would introduce a second, subtler leak.

3.2 Validation schemes

Scheme	Description	Models the task?
Random stratified 10-fold	rows shuffled, class ratio preserved per fold; averaged over 5 seeds	no
Time-ordered expanding window	5 blocks; fold i trains on blocks $1 \dots i$, tests on block $i+1$	yes
Chronological holdout	single split: first 70 % train, last 30 % test	yes

Table 3. Validation schemes compared in this study.

3.3 Metrics

Accuracy alone is uninformative here: predicting “no hazard” for every record already scores 0.9341. We therefore report ROC-AUC, PR-AUC (average precision), balanced accuracy, and recall/precision at a threshold that flags the riskiest 6.59% of shifts — the operating point implied by the base rate.

4 Results

4.1 Model $\times$ validation scheme

Model	Validation	ROC-AUC	PR-AUC	Accuracy	Bal. acc.	Recall
Logistic regression	Random 10-fold	0.755	0.195	0.904	0.610	0.271
Logistic regression	Time-ordered	0.668	0.084	0.927	0.555	0.148
Bagged CART	Random 10-fold	0.774	0.222	0.902	0.603	0.259
Bagged CART	Time-ordered	0.640	0.078	0.924	0.537	0.114

Table 4. Model $\times$ validation scheme. Random-CV figures are means over five seeds (standard deviations 0.003–0.008); time-ordered figures are deterministic.

Two observations follow. First, the **ranking of the models reverses** between schemes: under random CV the non-linear ensemble looks better, but under time-ordered validation it is worse. A more flexible model exploits

the leaked structure more effectively and collapses further when it is removed. Second, the loss is much larger in **PR-AUC** than in ROC-AUC. For a rare-event problem PR-AUC is the more relevant summary, so random CV overstates skill most in the metric that matters most.

PR-AUC by model and validation scheme

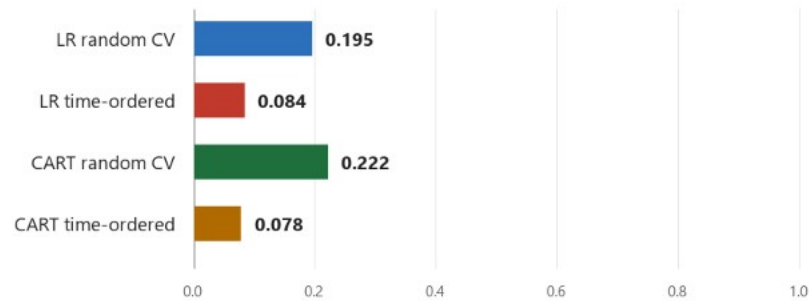


Figure 2. PR-AUC by model and validation scheme. The gap between the solid and adjacent bars is the optimism introduced by random cross-validation.

Precision-recall curves (out-of-fold predictions)

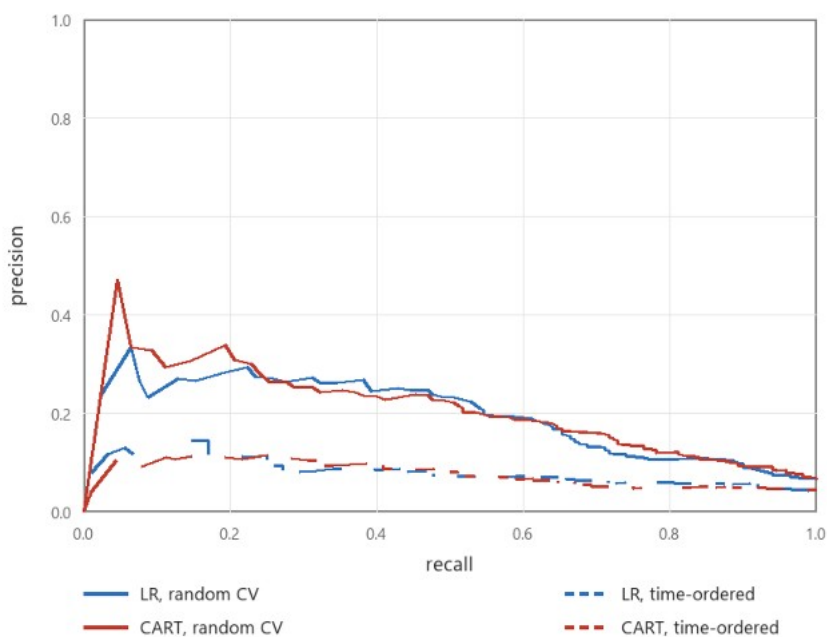


Figure 3. Precision-recall curves. Solid lines use random stratified 10-fold validation, dashed lines use time-ordered expanding-window validation. The vertical gap between a solid line and its dashed counterpart is the optimism introduced by random cross-validation.

ROC curves (out-of-fold predictions)

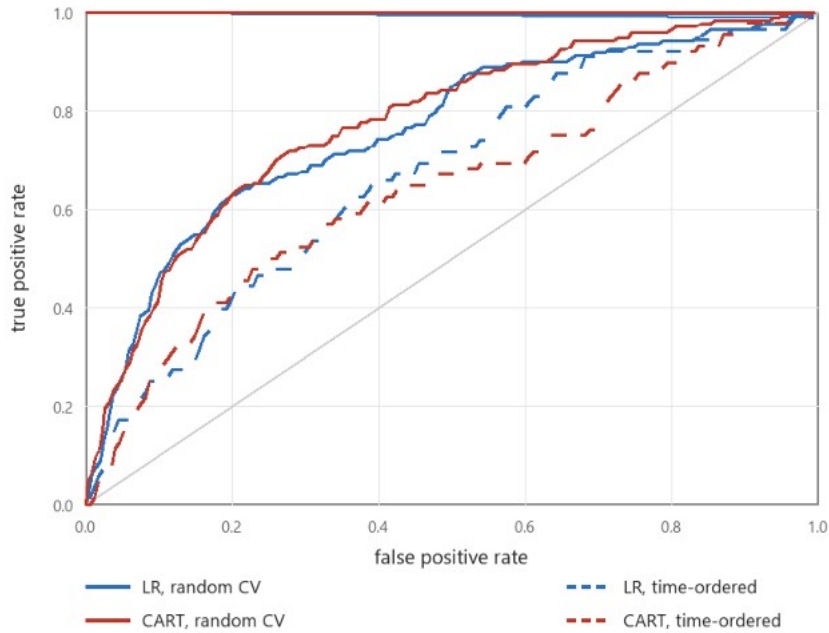


Figure 4. ROC curves for the same four combinations. The separation is far less visible here than in the precision-recall view, which is why PR-AUC is the more informative summary for this dataset.

The published baseline table for this dataset reports accuracy and balanced accuracy only, so a metric-for-metric comparison is not possible. Balanced accuracies there (50–69 %) bracket those obtained here, which is consistent with our models being of comparable quality rather than better.

4.2 Leakage probe

The record index, scaled to $[0, 1]$, is appended as an extra feature. It carries no physical information.

Model	Validation	ROC-AUC	PR-AUC
Logistic regression	Random 10-fold	0.762	0.206
Logistic regression	Time-ordered	0.609	0.063
Bagged CART	Random 10-fold	0.804	0.281
Bagged CART	Time-ordered	0.669	0.085

Table 5. Leakage probe: the record index added as a feature.

For the bagged CART ensemble the probe *raises* random-CV PR-AUC from 0.222 to 0.281 while leaving honest temporal performance essentially unchanged. This is the diagnostic signature of temporal leakage and is the clearest single piece of evidence in this study.

Univariate reference. Using the negated record index alone as a score gives ROC-AUC **0.670**. That single, physically meaningless variable matches the full 17-feature model’s honest time-ordered performance (0.668).

4.3 Chronological holdout

Model	n test	ROC-AUC	PR-AUC	Accuracy	Recall	Precision
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Logistic regression	774	0.568	0.090	0.941	0.115	0.115
Bagged CART	774	0.612	0.087	0.941	0.115	0.115

Table 6. Chronological holdout: train on the first 70 % of records, test on the last 30 %.

On a genuine future holdout the models are close to uninformative. The high accuracy is achieved by predicting “no hazard” almost always, which is exactly the failure mode that aggregate accuracy conceals. This negative result is reported deliberately: it is the honest estimate of what such a model would do in service.

4.4 Feature importance

Rank	Feature	Drop in ROC-AUC	s.d.
1	nbumps2	0.0645	0.0051
2	gpuls	0.0562	0.0040
3	energy	0.0522	0.0043
4	genergy	0.0431	0.0027
5	gdenergy	0.0291	0.0021
6	gdpuls	0.0219	0.0031
7	shift	0.0099	0.0018
8	nbumps3	0.0094	0.0018

Table 7. Permutation importance (drop in ROC-AUC when a column is shuffled), bagged CART fitted on all records, 5 repeats.

The signal is dominated by low-energy bump counts and by the previous shift’s geophone energy and pulse count. The categorical hazard assessments contribute little on top of the continuous monitoring channels — an observation worth re-checking with a model that incorporates explicit mechanism constraints, since permutation importance on correlated features understates the contribution of each member of a correlated group.

4.5 Probability calibration

Class weights are what make these models usable on a 6.6 %-positive dataset, but they have a side effect: the fit behaves as if the base rate were 0.50, so the output probabilities sit on the wrong scale. Ranking metrics are blind to this, because they depend only on the order of the scores. For an early-warning system, however, the difference between “8 % chance” and “33 % chance” is the difference between a usable threshold rule and a useless one.

Model	Variant	Brier	Skill	ECE	MCE	Mean predicted	Observed
LR	as fitted (class-weighted)	0.1462	-2.579	0.2904	0.5701	0.3331	0.0427
LR	after prior correction	0.0414	-0.013	0.0144	0.0413	0.0436	0.0427
CART	as fitted (class-weighted)	0.0433	-0.061	0.0374	0.1219	0.0662	0.0427
CART	after prior correction	0.0420	-0.028	0.0372	0.0935	0.0055	0.0427

Table 8. Probability calibration on the time-ordered out-of-fold predictions. “Skill” is the Brier skill score against always predicting the base rate; ECE is the expected calibration error.

The logistic regression is severely over-confident as fitted: its mean predicted probability is 0.333 against an observed rate of 0.0427, and its ECE is 0.290. The standard prior correction — a shift of the log-odds by $\log(\pi/(1-\pi)) - \log(0.5)$ — reduces ECE to 0.0144, a 95% reduction, and brings the mean prediction to 0.0436 against an observed 0.0427.

The bagged CART ensemble is far better calibrated even before correction (mean predicted 0.0662), because averaging leaf proportions over bootstrap samples already pulls the probabilities back towards the empirical rate. Here the same correction *over-corrects*: the mean prediction falls to 0.0055, now well below the observed 0.0427, although Brier score and maximum calibration error both improve slightly. The correction assumes the model was fitted at a prevalence of exactly 0.5; that assumption holds for the penalised linear fit but only approximately for the tree ensemble.

Brier skill scores remain at or below zero after correction for both models. Their probabilities are therefore no better than always quoting the base rate under a squared-error loss, even though their discrimination (ROC-AUC 0.668) is real. This is a concrete illustration of the distinction between discrimination and calibration: the model is useful for ranking and triage, but not for stating probabilities.

Logistic regression: probability calibration

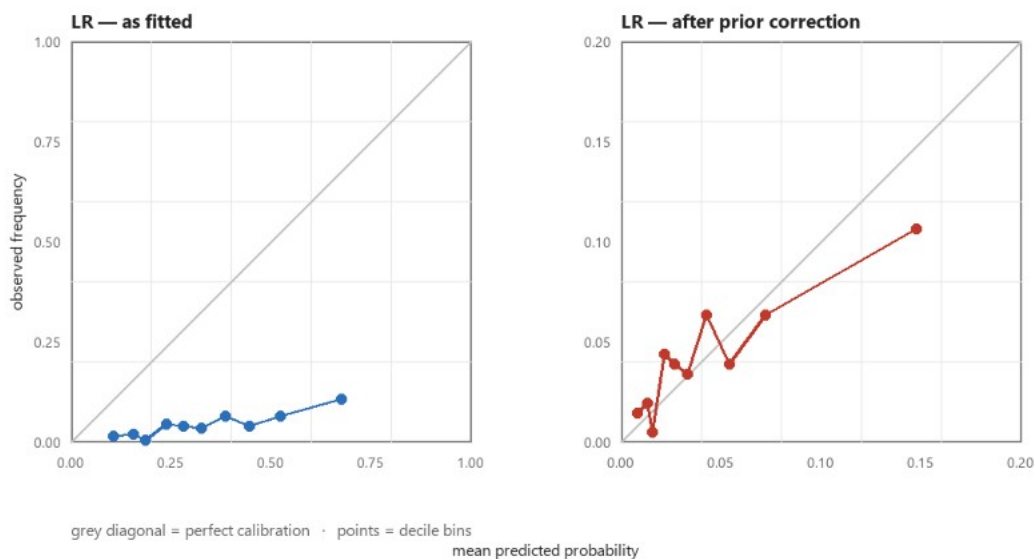


Figure 5. Reliability diagrams for the logistic regression, before and after prior correction. Points on the grey diagonal are perfectly calibrated. Left: the fitted model is over-confident across the whole range. Right: after a single log-odds shift the curve tracks the diagonal closely.

5 Discussion

1. Random cross-validation is optimistic on this dataset, and the optimism is model-independent. The effect appears in both a linear and a non-linear model, and is larger for the more flexible one.

2. Reported skill is highly sensitive to the validation protocol. The same model spans “useful” to “barely better than chance” depending only on how the data are split. Since the operational purpose is early warning, the honest number is the lower one.

3. A rank-deficient design matrix was present in the original feature set. Three of the 18 columns are constant and three linear dependencies exist among the bump-range counts. This does not damage tree-based models but makes unregularised logistic coefficients uninterpretable.

4. The drift structure is itself informative. A $7.5\times$ change in base rate across the record suggests the sequence spans distinct mining conditions. A model that appears to generalise across them under random CV may simply be memorising which regime each row belongs to.

5. Discrimination and calibration are separate properties, and only the first is usually reported. Class weighting buys discrimination on a rare-event problem at the cost of the probability scale: the logistic regression over-states risk by roughly a factor of eight until the log-odds are corrected. Since the operational use of an early-warning model is a threshold decision, the probability scale is not a detail. Reporting ROC-AUC or PR-AUC alone hides it.

6 Limitations

- The dataset has no explicit timestamp. Row order is treated as the chronological order of shifts, following the source documentation. If rows are instead grouped by longwall, the same argument applies with “group” in place of “time”: a random split still violates the independence assumption, so the conclusion holds either way, though the magnitude may differ.
- Two longwalls are present but cannot be separated, as no wall identifier is provided.
- No hyper-parameter search was performed. Tuning was deliberately avoided so that the comparison between validation schemes is not confounded by model selection.
- The published baselines report accuracy and balanced accuracy only, so comparison here is qualitative.
- Permutation importance on correlated features understates individual contributions.

7 Reproducibility

The full pipeline is available at the repository named on the title page and runs in under five minutes on a laptop CPU with no GPU:

```
python run_experiments.py
python verify_results.py
```

`verify_results.py` re-derives the dataset facts from the raw CSV and re-checks every quantity reported above; it exits non-zero if anything disagrees. It is not a formality: changing a single label in the input makes it fail.

The pipeline was executed on two very different stacks — Python 3.7.0 with numpy 1.21.6 and pandas 1.1.5, and Python 3.12 with numpy 2.3.x and pandas 3.0.x — and all reported quantities agreed to four decimal

places. The difference between random and time-ordered validation is therefore a property of the data and the protocol, not of a library version or a random-number stream.

References

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